REV A LAYOUT REVIEW. NOT FOR FABRICATION.

Four-slot carrier: protected external 5 V / 4 A, isolated GPIO

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM_FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.

J1
Pico 2 with headers, SC1632
GP0 1
GP1 2
GND 3
GP2 4
GP3 5
GP4 6
GP5 7
GND 8
GP6 9
GP7 10
GP8 11
GP9 12
GND 13
GP10 14
GP11 15
GP12 16
GP13 17
GP14 18
GP15 19
GP16 20
GP17 21
GND 22
GP18 23
GP19 24
GP20 25
GP21 26
GND 27
GP22 28
RUN 29
GP26_ADC0 30
3V3_OUT 31
3V3_EN 32
VSYS 33
VBUS 34
GP16 GP_EXP_RESET_N
GP17 P_IO_CE_N
GND GND
GP18 P_SPI_SCK
GP19 P_SPI_MOSI
GP20 P_EXP_CS_N
GP21 P_IO_CS_N
GND GND
GP22 P_S2_CS_N
GP26 P_S1_CS_N
3V3_OUT PICO_3V3
3V3_EN NC
VSYS PICO_VSYS
VBUS USB_VBUS

J3
SSQ-120-01-G-S
P_S1_IRQ 1 2 S1_5V
P_S1_BUSY 3 4 GND
GND 5 6 P_S2_IRQ
P_S2_IRQ 7 8 P_S2_BUSY
P_S2_BUSY 9 10 GND
P_S2_CS_N 11 12 P_S1_IRQ
P_S1_BUSY 13 14 P_S1_BUSY
GND 15 16 P_S1_CS_N
P_SPI_SCK 17 18 GND
P_SPI_MOSI 19 20 P_EXP_CS_N

J11
SSQ-120-01-G-S
P_EXP_RESET_N 1 2 GND
P_IO_CE_N 3 4 GND
GND 5 6 NC
7 8 NC
9 10 PICO_3V3
11 12 NC
13 14 NC
15 16 NC
17 18 GND
GND 19 20 GND
NC 21 22 PICO_VSYS
23 24 USB_VBUS

J9
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J2
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 IOREF_3V3
S1_RX_EN 7 8 GND

J10
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S2_CS_N
RADIO_5V 5 6 S1_CS_N
GND 7 8 S2_RESET_N
SPI_SCK 9 10 GND

J4
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
5 6 NC
7 8 IOREF_3V3
9 10 GND

J5
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J7
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J6
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 IOREF_3V3
S3_RX_EN 7 8 GND

J8
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

U5
SN74LVC8T245PWR
PICO_3V3 1 VCCA VCCB 20 GND
PICO_3V3 2 DIR OE 19 EXP_RESET_N
P_SPI_SCK 3 A1 B1 18 EXP_CS_N
P_SPI_MOSI 4 A2 B2 17 S4_CS_N
P_IO_CS_N 5 A3 B3 16 S3_CS_N
P_S2_CS_N 6 A4 B4 15 S2_CS_N
P_S1_CS_N 7 A5 B5 14 S1_CS_N
P_S4_CS_N 8 A6 B6 13 SPI_MISO_SBC
P_EXP_CS_N 9 A7 B7 12 SPI_SCK_SBC
P_EXP_MOSI_SBC 10 A8 B8 11 IOREF_3V3
GND 11 GND GND 10 IOREF_3V3
GND 12 GND VCCB 25 IOREF_3V3

U6
SN74LVC8T245PWR
PICO_3V3 1 VCCA VCCB 20 GND
GND 2 DIR OE 19 S4_IRQ
P_S1_BUSY 3 A1 B1 18 S4_BUSY
P_S1_IRQ 4 A2 B2 17 S3_IRQ
P_S2_BUSY 5 A3 B3 16 S3_BUSY
P_S2_IRQ 6 A4 B4 15 S2_IRQ
P_S3_BUSY 7 A5 B5 14 S2_BUSY
P_S3_IRQ 8 A6 B6 13 S1_IRQ
P_S4_BUSY 9 A7 B7 12 S1_BUSY
P_EXP_MOSI_SBC 10 A8 B8 11 IOREF_3V3
GND 11 GND GND 10 IOREF_3V3
GND 12 GND VCCB 25 IOREF_3V3

U7
SN74LVC1T45DBVR
PICO_3V3 1 VCCA VCCB 6 SPI_MISO
GND 2 GND 5 GND
P_SPI_MOSI 3 A B 4 IOREF_3V3

J12
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S2_RESET_N
SPI_SCK 9 10 GND

J13
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
5 6 NC
7 8 IOREF_3V3
9 10 GND

J14
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S6_CS_N
RADIO_5V 5 6 S5_CS_N
GND 7 8 S6_RESET_N
SPI_SCK 9 10 GND

J15
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 IOREF_3V3
S6_RX_EN 7 8 GND

J16
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 IOREF_3V3
S5_RX_EN 7 8 GND

J17
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J18
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J19
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 IOREF_3V3
S5_RX_EN 7 8 GND

J20
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J21
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S4_RESET_N
SPI_SCK 9 10 GND

J22
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J23
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 IOREF_3V3
S3_RX_EN 7 8 GND

J24
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J25
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 IOREF_3V3
S1_RX_EN 7 8 GND

J26
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J27
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J28
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J29
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J30
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J31
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J32
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J33
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J34
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J35
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J36
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J37
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J38
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J39
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J40
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J41
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J42
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J43
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J44
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J45
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J46
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J47
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J48
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J49
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J50
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J51
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J52
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J53
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J54
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J55
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J56
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J57
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J58
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J59
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J60
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J61
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J62
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J63
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J64
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J65
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J66
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J67
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J68
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J69
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J70
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J71
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J72
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J73
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J74
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J75
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J76
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J77
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J78
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J79
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J80
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J81
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J82
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J83
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J84
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J85
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J86
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J87
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J88
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J89
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J90
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J91
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J92
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J93
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J94
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J95
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J96
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J97
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J98
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J99
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J100
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J101
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J102
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J103
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J104
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J105
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J106
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J107
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J108
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J109
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J110
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J111
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J112
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J113
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J114
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J115
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J116
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J117
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J118
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J119
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J120
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J121
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J122
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J123
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J124
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J125
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J126
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J127
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J128
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J129
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J130
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J131
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J132
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J133
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J134
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J135
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J136
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J137
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J138
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J139
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J140
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J141
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J142
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J143
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J144
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J145
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J146
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J147
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S3_CS_N
RADIO_5V 5 6 S4_CS_N
GND 7 8 S3_RESET_N
SPI_SCK 9 10 GND

J148
SSQ-108-01-G-S
S3_BUSY 1 2 NC
S3_IRQ 3 4 NC
S3_TX_EN 5 6 NC
S3_RX_EN 7 8 IOREF_3V3
9 10 GND

J149
SSQ-108-01-G-S
S4_BUSY 1 2 NC
S4_IRQ 3 4 NC
S4_TX_EN 5 6 NC
S4_RX_EN 7 8 IOREF_3V3
9 10 GND

J150
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S5_CS_N
RADIO_5V 5 6 S6_CS_N
GND 7 8 S5_RESET_N
SPI_SCK 9 10 GND

J151
SSQ-108-01-G-S
S5_BUSY 1 2 NC
S5_IRQ 3 4 NC
S5_TX_EN 5 6 NC
S5_RX_EN 7 8 IOREF_3V3
9 10 GND

J152
SSQ-108-01-G-S
S6_BUSY 1 2 NC
S6_IRQ 3 4 NC
S6_TX_EN 5 6 NC
S6_RX_EN 7 8 IOREF_3V3
9 10 GND

J153
SSQ-110-01-G-S
GND 1 2 SPI_MISO
RADIO_5V 3 4 S1_CS_N
RADIO_5V 5 6 S2_CS_N
GND 7 8 S1_RESET_N
SPI_SCK 9 10 GND

J154
SSQ-108-01-G-S
S1_BUSY 1 2 NC
S1_IRQ 3 4 NC
S1_TX_EN 5 6 NC
S1_RX_EN 7 8 IOREF_3V3
9 10 GND

J155
SSQ-108-01-G-S
S2_BUSY 1 2 NC
S2_IRQ 3 4 NC
S2_TX_EN 5 6 NC
S2_RX_EN 7 8 IOREF_3V3
9 10 GND

J156
SS

REV A LAYOUT REVIEW. NOT FOR FABRICATION.

915 MHz LoRa card

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.

J1
TSW-110-07-G-S
GND 1
SLOT_5V 2
SLOT_5V 3
GND 4
SPI_SCK 5
6 SPI_MOSI
7 SPI_MISO
8 CS_N
9 RESET_N
10 GND

J2
TSW-108-07-G-S
BUSY 1
IRQ 2
TX_EN 3
RX_EN 4
5 AUX0
6
7 IOREF_3V3
8 GND

U1
E22-900M30S
GND 1
GND 2
GND 3
GND 4
GND 5
RX_EN 6
TX_EN 7
AUX0 8
SLOT_5V 9
SLOT_5V 10
GND 11
12 GND
13 IRQ
14 BUSY
15 RESET_N
16 SPI_MISO_SRC
17 SPI_MOSI
18 SPI_SCK
19 CS_N
20 GND
21
22 GND

Module pad 21 ANT: NC, use module IPEX. Both pads 9 and 10 are VCC. H1 is the mandatory alignment-post hole.
DIO2 is AUX0 for a local test point only: do not connect it to a Pico GPIO.
AUX1 is reserved. DIO3 is internal TCXO control, configured to 2.2 V.

R5
22
SPI_MISO_SRC 1
2 SPI_MISO

TP1
SPI_SCK
SPI_SCK 1
PROBE

TP2
SPI_MOSI
SPI_MOSI 1
PROBE

TP3
SPI_MISO
SPI_MISO 1
PROBE

TP4
CS_N
CS_N 1
PROBE

#FLG01
POWER DECLARATION: SLOT_5V
SLOT_5V 1
POWER

#FLG02
POWER DECLARATION: GND
GND 1
POWER

#FLG03
POWER DECLARATION: IOREF_3V3
IOREF_3V3 1
POWER

TP5
RESET_N
RESET_N 1
PROBE

TP6
BUSY
BUSY 1
PROBE

C1
220uF
SLOT_5V 1
+
-
2 GND

C2
22uF
SLOT_5V 1
1
2
2 GND

C3
100nF
SLOT_5V 1
1
2
2 GND

TP7
IRQ
IRQ 1
PROBE

TP8
TX_EN
TX_EN 1
PROBE

TP9
RX_EN
RX_EN 1
PROBE

TP10
IN_5V
SLOT_5V 1
PROBE

R1
10k
IOREF_3V3 1
1
2
2 CS_N

R2
10k
IOREF_3V3 1
1
2
2 RESET_N

TP11
CLIP_GND
GND 1
PROBE

TP12
AUX0
AUX0 1
PROBE

R3
10k
TX_EN 1
1
2
2 GND

R4
10k
RX_EN 1
1
2
2 GND

H1
M3 alignment hole

TP13
MOD_5V
SLOT_5V 1
PROBE

TP14
MOD_GND
GND 1
PROBE

TP15
IOREF
IOREF_3V3 1
PROBE

MISO: source -> 22R -> slot A7. Fit OR option: scope/IBIS verify.
Place at module output/socket, before shared bus. No long stubs.

KEYED SLOT: H1 engages the carrier M3 x 25 post before electrical contact.
All headers and passives factory fitted. No user soldering or antenna-selector rework.
Power declarations describe power arriving through the slot and buck output after L1.
Layout authorized; no fabrication release. See DESIGN_REVIEW.md.

SCHEMATIC REVIEW — NOT FOR FABRICATION

Sheet: /
File: e22-900m30s-rev-a.kicad_sch

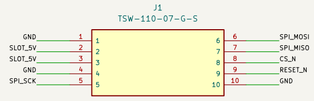
Title: 915 MHz LoRa card

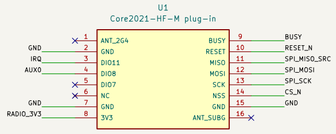
Size: A2	Date: 2026-09-17	**Rev: A-schematic**
KiCad E.D.A. 10.0.6		Id: 1/1

REV A LAYOUT REVIEW. NOT FOR FABRICATION.

915 MHz FLRC socket adapter

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.





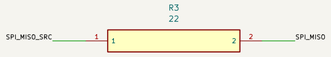








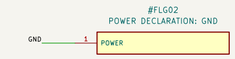
U1 represents the plug-in module, not a soldered module footprint.
Socket contacts for module pins 1, 5, 6 and 16 are isolated.
TX_EN/RX_EN slot contacts unused. No external RF switch added.
J3/J4: 15.24 mm row spacing; top/bottom numbering in FOOTPRINT_REVIEW.md.





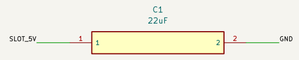


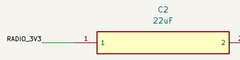


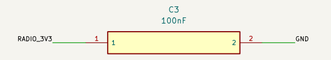


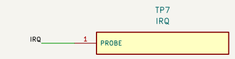




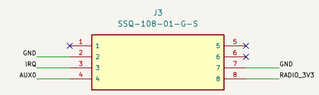


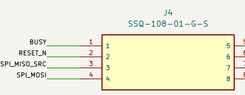


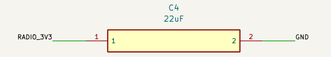






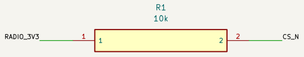


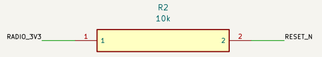
























MISO: source -> 22R -> slot A7. Fit OR option: scope/IBIS verify.
Place at module output/socket, before shared bus. No long stubs.

KEYED SLOT: H1 engages the carrier M3 x 25 post before electrical contact.
All headers and passives factory fitted. No user soldering or antenna-selector rework.
Power declarations describe power arriving through the slot and buck output after L1.
Layout authorized; no fabrication release. See DESIGN_REVIEW.md.

SCHEMATIC REVIEW — NOT FOR FABRICATION

Sheet: /
File: core2021-hf-m-rev-a.kicad_sch

Title: 915 MHz FLRC socket adapter

Size: A2	Date: 2026-09-17	**Rev: A-schematic**
KiCad E.D.A. 10.0.6		Id: 1/1

REV A LAYOUT REVIEW. NOT FOR FABRICATION.

2.4 GHz LoRa or FLRC card, same hardware

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.

J1
TSW-110-07-G-S
GND 1
SLOT_5V 2
SLOT_5V 3
GND 4
SPI_SCK 5
6 SPI_MOSI
7 SPI_MISO
8 CS_N
9 RESET_N
10 GND

J2
TSW-108-07-G-S
BUSY 1
IRQ 2
TX_EN 3
RX_EN 4
5 AUX0
6
7
8 GND

U1
E28-2G4M27S
RADIO_3V3 1 VCC
GND 2 GND
SPI_MISO_SRC 3 MISO_TX
SPI_MOSI 4 MOSI_RX
SPI_SCK 5 SCK_RTSN
CS_N 6 NSS_CTS
GND 7 GND
RX_EN 8 RX_EN
9 TX_EN
10 GND
11 RESET_N
12 BUSY
13 IRQ
14 AUX0
15
16 GND

U2
AP63203WU-7
RADIO_3V3 1 FB
SLOT_5V 2 EN
SLOT_5V 3 VIN
4 GND
5 SW
6 BST

R5
22
SPI_MISO_SRC 1 — 2 SPI_MISO

ASSEMBLY VARIANT: IPEX ONLY. Selector MUST match manual section 9.2 right image.
MCP23517 controls TX_EN/RX_EN before RF commands. No automatic RX/TX modes.
Buck PCB layout must follow figure 25, 2 oz copper on both layers.

#FLG01
POWER DECLARATION: SLOT_5V
SLOT_5V 1 POWER

#FLG02
POWER DECLARATION: GND
GND 1 POWER

#FLG03
POWER DECLARATION: RADIO_3V3
RADIO_3V3 1 POWER

L1
4.7uH
SW 1 — 2 RADIO_3V3

C1
22uF
SLOT_5V 1 — 2 GND

C2
22uF
RADIO_3V3 1 — 2 GND

C3
22uF
RADIO_3V3 1 — 2 GND

C4
100nF
BST 1 — 2 SW

C5
220uF
SLOT_5V 1 + — 2 GND

C7
22uF
RADIO_3V3 1 — 2 GND

C8
22uF
SLOT_5V 1 — 2 GND

C6
100nF
RADIO_3V3 1 — 2 GND

R1
10k
RADIO_3V3 1 — 2 CS_N

R2
10k
RADIO_3V3 1 — 2 RESET_N

R3
10k
TX_EN 1 — 2 GND

R4
10k
RX_EN 1 — 2 GND

H1
M3 alignment hole

MISO: source -> 22R -> slot A7. Fit OR option: scope/IBIS verify.
Place at module output/socket, before shared bus. No long stubs.

KEYED SLOT: H1 engages the carrier M3 x 25 post before electrical contact.
All headers and passives factory fitted. No user soldering or antenna-selector rework.
Power declarations describe power arriving through the slot and buck output after L1.
Layout authorized; no fabrication release. See DESIGN_REVIEW.md.

TP1
SPI_SCK
SPI_SCK 1 PROBE

TP2
SPI_MOSI
SPI_MOSI 1 PROBE

TP3
SPI_MISO
SPI_MISO 1 PROBE

TP4
CS_N
CS_N 1 PROBE

TP5
RESET_N
RESET_N 1 PROBE

TP6
BUSY
BUSY 1 PROBE

TP7
IRQ
IRQ 1 PROBE

TP8
TX_EN
TX_EN 1 PROBE

TP9
RX_EN
RX_EN 1 PROBE

TP10
IN_5V
SLOT_5V 1 PROBE

TP11
CLIP_GND
GND 1 PROBE

TP12
AUX0
AUX0 1 PROBE

TP13
OUT_3V3
RADIO_3V3 1 PROBE

TP14
REG_GND
GND 1 PROBE

TP15
EN_5V
SLOT_5V 1 PROBE

TP16
SW
SW 1 PROBE

TP17
FB
RADIO_3V3 1 PROBE

TP18
MOD_3V3
RADIO_3V3 1 PROBE

TP19
MOD_GND
GND 1 PROBE

TP20
SW_GND
GND 1 PROBE

SCHEMATIC REVIEW — NOT FOR FABRICATION

Sheet: /
File: e28-2g4m27s-rev-a.kicad_sch

Title: 2.4 GHz LoRa or FLRC card, same hardware

Size: A2	Date: 2026-09-17	**Rev: A-schematic**
KiCad E.D.A. 10.0.6		Id: 1/1